

Vishal's Architecture Insights & Recommendations

Project

Automated Photo Identification and Classification for Traffic Violations Using Computer Vision

Objective

The objective is not only to detect traffic violations but to build a **production-ready, scalable, explainable, and enforcement-focused AI platform** that can be deployed across cities and integrated with traffic management systems.

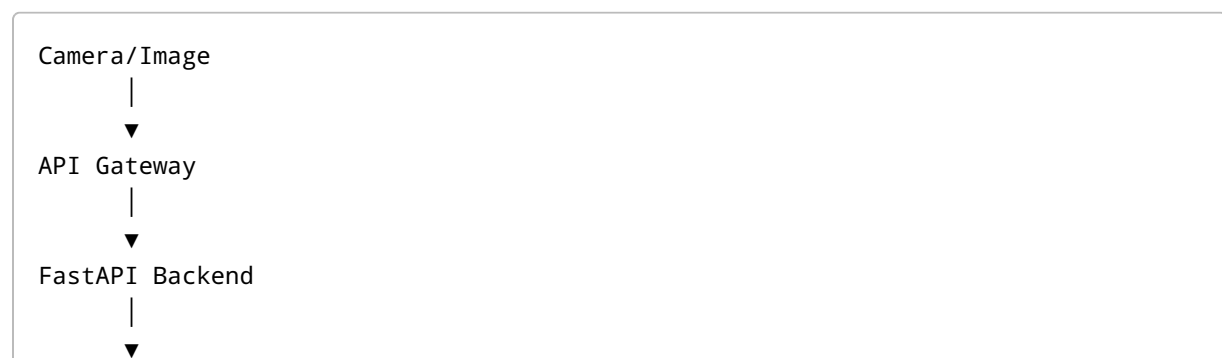
My Key Contributions

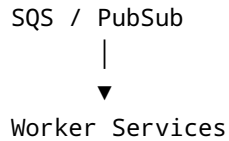
1. Scalable Cloud Architecture
 2. Queue-Based Processing Pipeline
 3. Explainable AI Layer
 4. Confidence-Based Enforcement
 5. Automated Evidence Generation
 6. Analytics & Intelligence Layer
 7. Cloud Deployment Strategy (AWS/GCP)
 8. Future Production Scaling using Kubernetes
-

1. Scalable Processing Architecture

Instead of processing requests synchronously, the entire platform should be event-driven and queue-based.

Proposed Flow





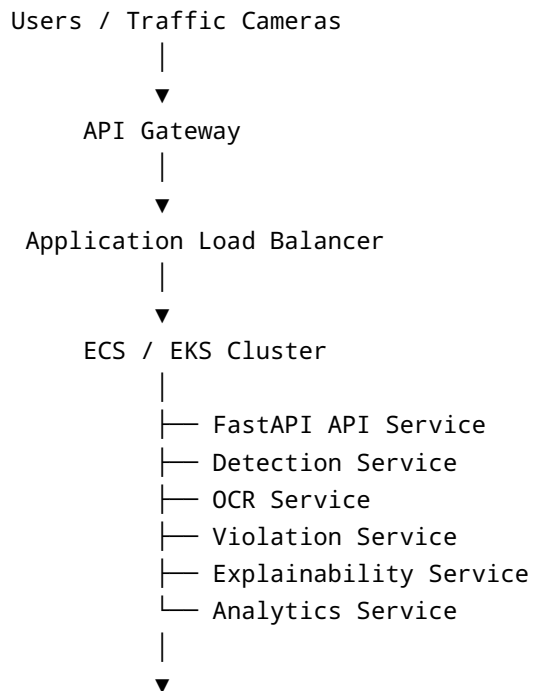
Worker Services

- Enhancement Service
- Detection Service
- OCR Service
- Violation Service
- Explainability Service
- Analytics Service

Benefits

- Handles thousands of images per minute
- Prevents API bottlenecks
- Supports horizontal scaling
- Fault tolerant architecture
- Cloud native deployment

2. Cloud Architecture (High Level)





3. Explainable AI Layer

Traditional systems only detect violations.

Our system should also explain WHY the violation occurred.

Example Model Output

```
{
  "signal": "red",
  "vehicle_crossed_line": true,
  "distance_crossed": "1.8m",
  "confidence": "96%"
}
```

Generated Explanation

The vehicle crossed the designated stop line while the traffic signal was red.

Vehicle crossed the stop line by 1.8 meters.

Confidence Score: 96%.

Violation Type:
Red Light Violation.

Benefits

- Transparency
- Trustworthy enforcement
- Easier review by traffic authorities
- Better legal defensibility

4. Confidence-Based Enforcement Workflow

To reduce false challans and improve reliability:

Violation Engine
|
▼
Confidence Audit

Decision Logic

Confidence > 95%
|
▼
Auto Challan

Confidence 70% - 95%
|
▼
Human Review Queue

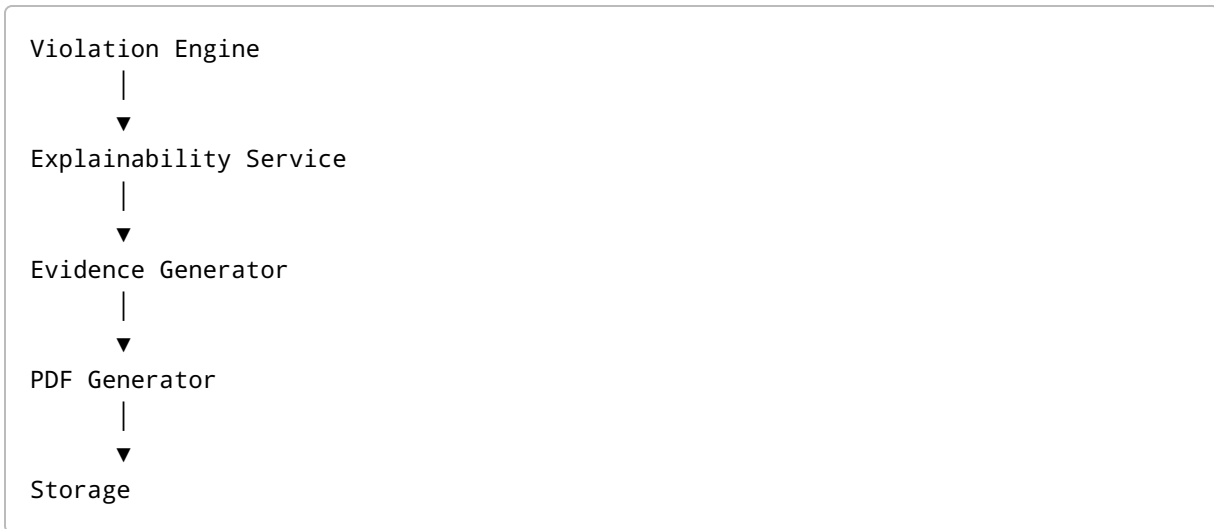
Confidence < 70%
|
▼
Rejected / Reprocessing

Benefits

- Reduces false positives
- Human-in-the-loop validation
- Production-ready enforcement workflow

5. Automated Evidence Generation

After violation detection:



Generated Evidence Packet

- Annotated Violation Image
- Vehicle Number
- Timestamp
- Location
- Violation Type
- Confidence Score
- Generated Explanation
- PDF Report

Benefits

Creates a complete digital evidence package ready for enforcement.

6. Analytics & Intelligence Layer

The platform should provide insights, not only detection.

Features

Violation Heatmaps

Identify locations with high violation frequency.

Repeat Offender Detection

Example:

Vehicle:
MH12AB1234

Total Violations:
8

Risk Score:
HIGH

Violation Trends

- Daily trends
- Weekly trends
- Monthly trends

Zone-wise Analytics

- Area-wise violation statistics
- High-risk junction identification

7. Smart Police Deployment Recommendation

Additional insight:

If multiple violations occur repeatedly at the same location, the system should flag that location.

Example

Location A:
450 violations/week

Location B:
15 violations/week

Location C:
390 violations/week

Recommendation

Deploy Traffic Officers
at Location A and Location C

This allows authorities to use AI-generated heatmaps for operational planning and traffic management.

8. AWS Deployment Strategy

Frontend

- S3
- CloudFront

API Layer

- API Gateway

Backend

- ECS Fargate / EKS

Queue

- SQS

Event Routing

- EventBridge

Cache

- ElastiCache Redis

Database

- RDS PostgreSQL

Storage

- Amazon S3

Monitoring

- CloudWatch

AI Models

- SageMaker

- Bedrock

OCR

- Textract
- PaddleOCR Service

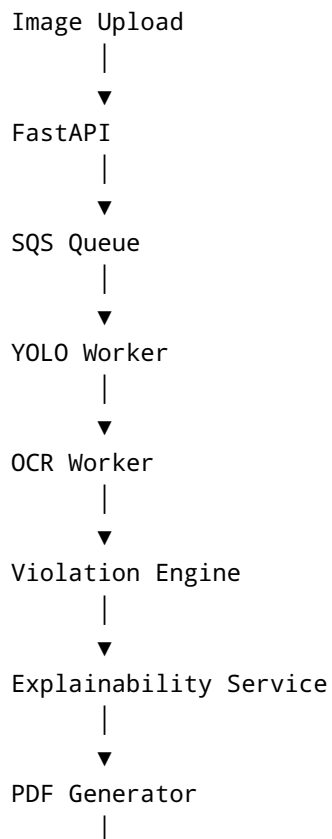
9. Deployment Option 1 (MVP)

ECS Fargate

Recommended for:

- Hackathons
- POCs
- MVP Deployments

Processing Flow



▼
S3 Storage

Benefits

- Lower cost
- Easy deployment
- Fully managed infrastructure

10. Deployment Option 2 (Production Scale)

EKS Kubernetes

Recommended for:

- City-wide deployments
- State-wide deployments
- High traffic systems

Architecture

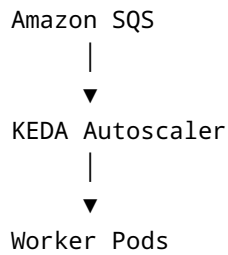
```
graph TD
    Internet --> AWS_ALB_Ingress[AWS ALB Ingress]
    AWS_ALB_Ingress --> EKS_Cluster[EKS Cluster]
```

Services

Namespace: traffic-ai

- FastAPI Pods
- Detection Pods
- OCR Pods
- Violation Pods
- Explainability Pods
- Analytics Pods

Autoscaling



Example:

100 Images
→ 2 Pods

10000 Images
→ 50 Pods

No manual scaling required.

11. Future Production Enhancements

- GPU Worker Nodes
- Edge AI Processing
- Real-Time CCTV Stream Analysis
- Smart City Integration
- Predictive Violation Analytics
- Multi-City Deployment
- Federated Learning
- Continuous Model Retraining

Final Vision

Our goal is to build more than a traffic violation detector.

We aim to build a complete AI-powered Traffic Enforcement Platform that provides:

- Automated Detection
- Explainable Decisions
- Human Review Workflow

- Automated Evidence Generation
- Smart Analytics
- Police Deployment Recommendations
- Cloud-Native Scalability
- Production-Ready Architecture